AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

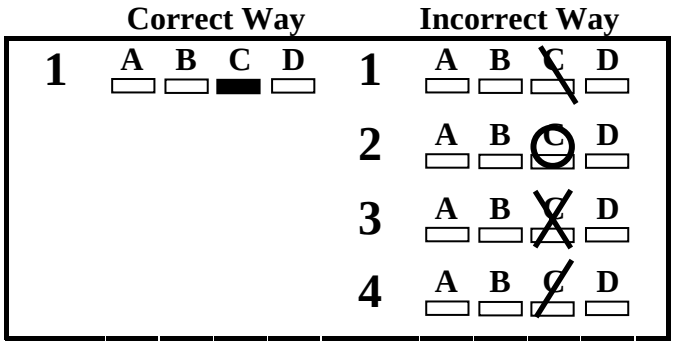
MAY 2011

Biology Paper I

Time allowed: 35 minutes Marks 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

1. The role of fungi in the production of various antibiotics is studied in

   A. taxonomy.
   B. morphology.
   C. paleontology.
   D. pharmacology.

2. The production of insulin on a large scale through bacteria has made insulin inexpensive and easily available to diabetic patients.

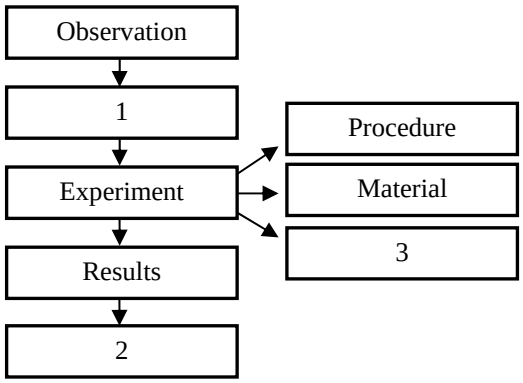
   The above information shows the relationship of biology with

   A. mathematics.
   B. economics.
   C. geography.
   D. physics.

3. Which of the following is the correct order of levels of organization in plants from simple to complex?

   A. Xylem, stem, chloroplast
   B. Chloroplast, xylem, stem
   C. Xylem, chloroplast, stem
   D. Chloroplast, stem, xylem

4. In solving a biological problem through a scientific method a number of steps are involved, some of these are shown below.



   Which of the following correctly identifies 1, 2 and 3?

	1	2	3
A	Hypothesis	Precautions	Conclusion
B	Conclusion	Precautions	Hypothesis
C	Hypothesis	Conclusion	Precautions
D	Conclusion	Hypothesis	Precautions

5. Which of the following characteristics of algae make them similar to fungi?

   A. Unicellular reproductive organs
   B. Capability of photosynthesis
   C. Non-septate hyphae
   D. Cellulose cell wall

6. Which of the following is correct about pteridophytes and spermatophytes?

   A. Vascular plants
   B. Seed producing
   C. Thallus body
   D. Gametophyte dominant

7. The animals included in class Reptilia and Amphibia are in a way similar because both

   A. live in water only.
   B. are cold blooded animals.
   C. have thick and dry skin.
   D. lay their eggs in water.

8. An invertebrate is found with the following characteristics.

   - Lives in sea water only.
   - Body is covered with spines.
   - Internal skeleton is present.

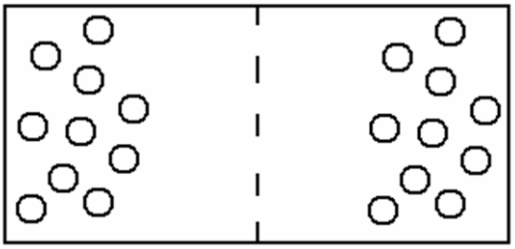
   In which phylum will it be placed?

   A. Mollusca
   B. Nematoda
   C. Coelenterata
   D. Echinodermata

9. A long protoplasmic extension is found in

   A. mesophyll cells.
   B. white blood cells.
   C. xylem vessels.
   D. root hair cells.

10. The diagram shows the molecules of two different substances A and B.



A B

Which option is correct about the phenomenon shown above?

	Difference in concentration	Movement of molecules
A	Zero	No movement
B	Higher at A	Towards B
C	Higher at B	Towards A
D	Zero	From A to B and B to A

11. Two different types of cells are placed in a hypotonic solution. Which of the following options is correct about the condition of these cells after some time?

	Red blood cells	Guard cells
A	Flaccid	Turgid
B	Turgid	Flaccid
C	Turgid	Turgid
D	Flaccid	Flaccid

12. Which of the following correctly identifies the examples of soft and hard connective tissues?

	Soft Connective Tissue	Hard Connective Tissue
A	Tendon	Cartilage
B	Cartilage	Bone
C	Fatty tissue	Tendon
D	Bone	Fatty tissue

13. Which phases of the cell cycle are collectively called interphase?

A. G_1, G_2 and M
B. S, G_2 and M
C. G_1, G_2 and S
D. G_1, M and S

14. The original number of chromosomes in daughter cells is maintained when the parent chromosomes undergo

    A. crossing over.
    B. condensation.
    C. cytokinesis.
    D. replication.

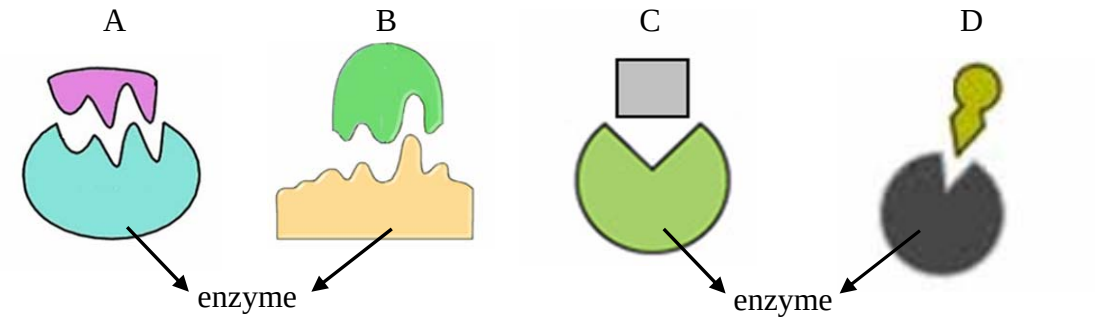
15. All of the following are advantages of mitosis EXCEPT

    A. asexual reproduction.
    B. genetic variation.
    C. cell replacement.
    D. genetic stability.

16. The conversion of insoluble fibrinogen into fibrin threads is carried out by enzyme thrombin which works in the presence of vitamin K only. Here, vitamin K is acting as a/an

    A. substrate.
    B. active site.
    C. co-factor.
    D. inhibitor.

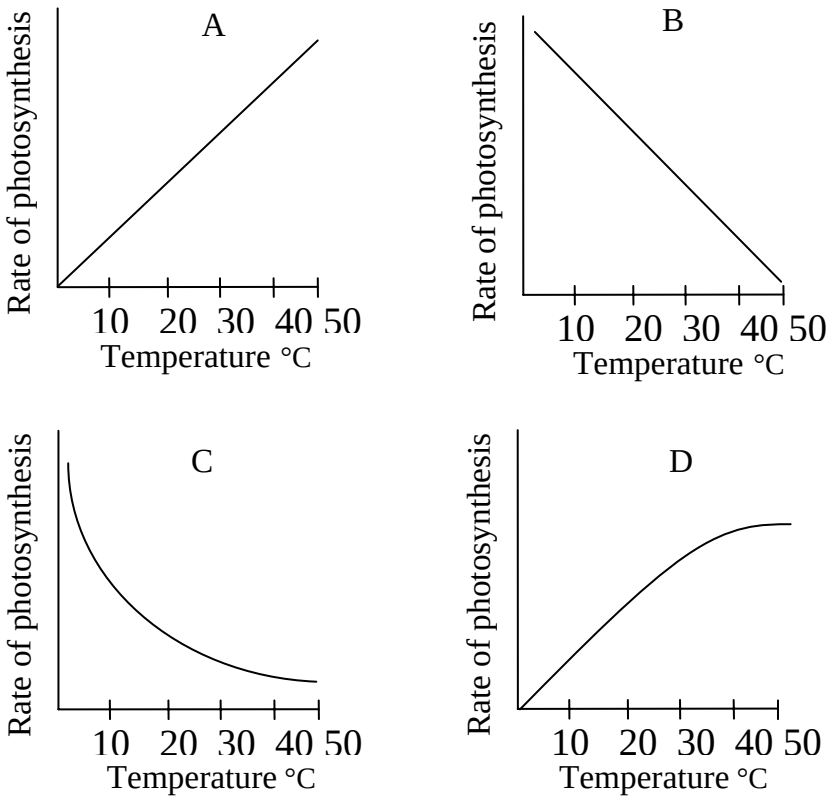
17. Which diagram shows denaturation of an enzyme?



18. The dark stage of photosynthesis is called dark because

    A. it is named after the scientist who discovered it.
    B. it requires a dark green pigment, chlorophyll.
    C. it is independent of light energy.
    D. it takes place only at night.

19. Which of the following graphs shows the effect of temperature on the rate of photosynthesis?



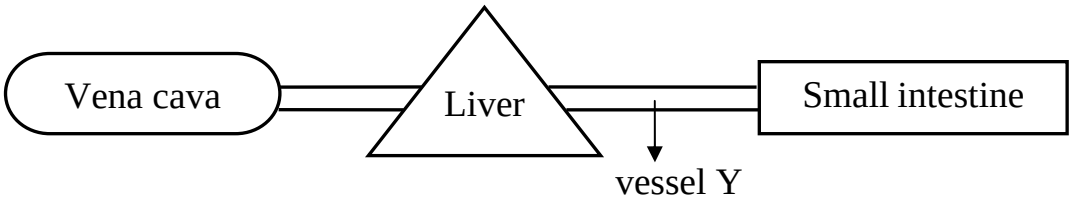
20. During aerobic respiration glucose is

A. hydrogenated.
B. hydrolysed.
C. oxidized.
D. reduced.

21. The polypeptides are completely digested into amino acids by the enzyme

A. amylase.
B. erepsin.
C. catalase.
D. lipase.

22. Which of the following is correct about vessel Y?



	Identification of Y	Composition of fluid inside Y
A	Hepatic vein	Glucose
B	Hepatic vein	Glycogen
C	Hepatic portal vein	Glucose
D	Hepatic portal vein	Glycogen

23. Which of the following identifies xylem vessels?

	Transport	Support	Absorption
A	✓	✓	×
B	✓	✓	✓
C	✓	×	×
D	×	✓	✓

24. White blood cells and platelets are similar in a way that both

A. are produced in the liver.
B. live for 120 days in blood.
C. are smaller than red blood cells.
D. protect the body against bacteria.

25. A person with blood group A can donate blood to

A. A only.
B. A and AB.
C. AB and O.
D. A and O.

END OF PAPER

Please use this page for rough work